SUMMER TERM 2022
CENTRALLY-MANAGED ONLINE EXAMINATION
ECON0039: ADVANCED MACROECONOMICS

Time allowance
You have 2 hours to complete this examination, plus an Upload Window of 20 minutes. The Upload Window is for uploading, completing the Cover Sheet and correcting any minor mistakes and should not be used for additional writing time.

If you have been granted SoRA extra time and/ or rest breaks, your individual examination duration will be extended pro-rata and you will also have the 20-minute Upload Window added to your individual duration.

All work must be submitted anonymously in a PDF file and you should follow the instructions for submitting an online examination in the AssessmentUCL Guidance for Students.

If you miss the submission deadline, you will not be able to submit your work via AssessmentUCL and you will not be permitted to submit the work via email or any other channel. If you are unable to submit your work due to technical difficulties which are substantial and beyond your control, you should apply for a Deferral via the AssessmentUCL Query Form.

Page Limit: No page limit

All questions should be answered.

Part A carries 1/3 of the total mark, Part B 1/3 and Part C 1/3. Questions B.1 and B.2 have the same weight.

If you have a query about the examination paper, instructions or rubric, you should complete an AssessmentUCL Query Form. Please note that you will not receive a response during your examination. By submitting this assessment, you are confirming that you have not violated UCL's Assessment Regulations relating to Academic Misconduct contained in Section 9 of Chapter 6 of the Academic Manual.

PART A : Demand and supply shocks in a "UV" model

THIS PROBLEM considers a general equilibrium model in the UV setting of Lecture 4 on Business Cycles. The economy features a representative firm and a representative household.

Technology : The representative firm produces consumption c and investment x using labour according to the technology

$$(c^\sigma + x^\sigma)^{\frac{1}{\sigma}} = A\ell^\alpha$$

with $0 < \alpha < 1$ and $\sigma \geq 1$. ℓ represents labour input.

Preferences : The representative household utility is of the UV form, and is given by

$$U(c, \ell) + V(x, \theta) = \gamma \ln c - B\ell + \theta \ln x,$$

where ℓ is labour.

Markets : All markets are competitive. The representative household owns the firm and receives its profits π. It is assumed that the consumption good is the numéraire (its price is normalized to one), while the price of investment is q and the wage is w.

The household budget constraint (that will always bind) is

$$c + qx \leq w\ell + \pi.$$

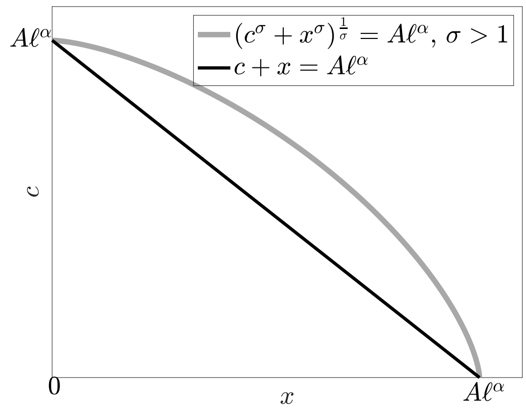
A.1 The profit maximization problem of the firm is

$$\max_{x,c,\ell} \pi = c + qx - w\ell$$

$$\text{subject to } (c^\sigma + x^\sigma)^{\frac{1}{\sigma}} = A\ell^\alpha$$

Figure A1 shows the production possibility frontier (ppf) of the economy for $\sigma = 1$ and $\sigma > 1$. The ppf is the locus of the couples (c, x) that can be produced for a given level of input labour ℓ.

Figure A1 - Production Possibility Frontier for $\sigma = 1$ and $\sigma > 1$



Explain why the slope of the ppf can be interpreted as the negative of the relative price q. Discuss the difference between the case $\sigma = 1$ and $\sigma > 1$.

A.2 From now on, we assume that σ is large enough, more precisely that $\sigma > \frac{1}{\alpha}$.

Show that profit maximisation can be rewritten as

$$\max_{x,c} \pi = c + qx - wA^{-\frac{1}{\alpha}} \left(c^\sigma + x^\sigma\right)^{\frac{1}{\alpha\sigma}}$$

and derive first order conditions to this maximization with respect to c, denoted equation (1), and x (eq. (2)).

A.3 Show that the utility maximisation problem of the household can be written as

$$\max_{x,\ell} \gamma \ln(w\ell + \pi - qx) - B\ell + \theta \ln x$$

Derive first order conditions to this maximisation, with respect to x (eq. (3)) and ℓ (eq. (4)). The budget constraint is denoted as equation (0).

A.4 Define the general equilibrium of this economy.

A.5 Solve for equilibrium prices and quantities.
Hint: (i) use (1) and (2) to find $\frac{x}{c}$ as a function of q (denoted equation (5)); (ii) use (5) and (3) to find q; (iii) derive a solution for $\frac{x}{c}$ (denoted equation (7)); (iv) take (1), rearrrange terms to have $\frac{x}{c}$ and replace it by its solution, use (4) to eliminate w and obtain the equilibrium value of c; (v) solve for x using the solution for $\frac{x}{c}$; (vi) use $(c^\sigma + x^\sigma)^{\frac{1}{\sigma}} = A\ell^\alpha$ to find ℓ.

A.6 Can an investment demand shock ($\partial\theta > 0$) create business cycles comovements? (Don't forget that $\sigma > \frac{1}{\alpha}$.) Explain.

PART B : Questions

WRITE A structured short essay for each of the two themes below. Each essay should be roughly 500 words.

B.1 The fiscal origin of hyperinflation.

B.2 Definition and measure of business cycles.

PART C : Discussion

BELOW IS an article published in The Economist in December 2021 and entitled "Japan's economy is stronger than many realise". As always in The Economist, the article is not signed. Read carefully this article before answering the questions below. For each question, you should propose a structured answer.

C.1 Write a 10-line summary.

C.2 Explain why the first quote of ADAM SMITH illustrates the functioning of an economy in the Malthusian regime (roughly 500 words).

C.3 Show where the text refers to what we have called in class a "liquidity trap" (to be defined) (roughly 500 words).

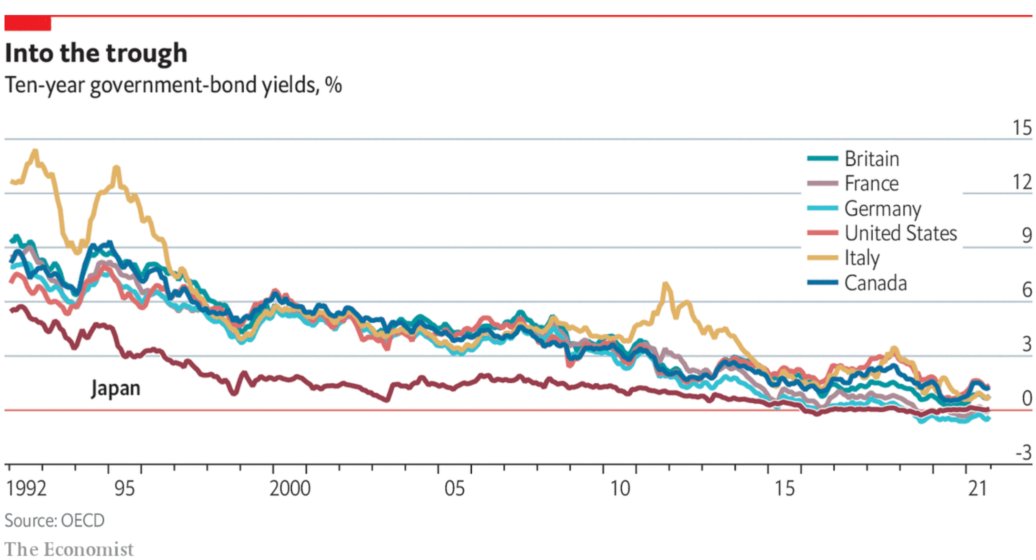
“Japan's economy is stronger than many realise.
Not bad, but could be better ”
The Economist, Dec 7th 2021

“The most decisive mark of the prosperity of any country is the increase of the number of its inhabitants,” wrote Adam Smith in “The Wealth of Nations” in 1776. Later David Ricardo and Thomas Malthus traded barbs over whether the food supply would keep up. By 1937 John Maynard Keynes was warning of future population decline, with deleterious economic effects.

Japan is the canary in this coal-mine. In the 1980s its booming economy struck fear in the world. After the bubble burst in the 1990s, public debt ballooned and deflation set in. Many in the West said Japan's debt was unsustainable and the Bank of Japan (BOJ) should do more to boost inflation. In 2013 the BOJ's governor, Kuroda Haruhiko, embarked on dramatic monetary easing. The debt hovered around 230% of GDP. A strange thing ensued: no fiscal crisis struck, nor did inflation come near the 2% target. “The standard textbook on macroeconomics needs an additional few chapters–it doesn't capture the problems Japan faced,” says Shirakawa Masaaki, Mr Kuroda's predecessor.

Many rich countries now face similar “secular stagnation”: low inflation, low interest rates and low growth. Although higher inflation has emerged recently, financial markets suggest secular stagnation will return soon. Demography is a big factor; Japan simply started ageing and shrinking earlier. As Japan has adapted and others have become more like it, some economists are seeing its economy in a new light.

Figure C.1



Debt has not turned out to be such a problem. "What we thought used to be fiscal limits are no longer fiscal limits," argues Adam Posen, of the Peterson Institute for International Economics (PIIE) think-tank. "[Japan] has forced people to confront reality: the interest rate can stay below the growth rate for very long periods." The public debt has been above 100% of GDP for almost 25 years without causing a crisis.

It helps that the country borrows in its own currency, the government has big financial assets and the BOJ holds a big share of the debt. But as David Weinstein of Columbia University argues, Japan has also managed to contain spending. Since 2000, he writes in a paper with Mark Greenan, spending per head on the elderly has actually fallen. "There's a quiet functionality that people miss," says Mr Weinstein. "Markets are sanguine because Japan has a rare ability to adjust." With low marginal tax rates, there is also room to raise revenues.

Some still fear what would happen if interest rates were to go up. The argument that Japan need not worry about its debt because it can respond by levying taxes is "too academic", says Yoshikawa Hiroshi, president of Rissho University. Raising consumption taxes is a political loser. Policymakers are also haunted by the spectre of external shocks that create new fiscal needs. Recently Yano Koji, a vice-minister of finance, caused a stir with a column comparing the country's fiscal situation to the Titanic.

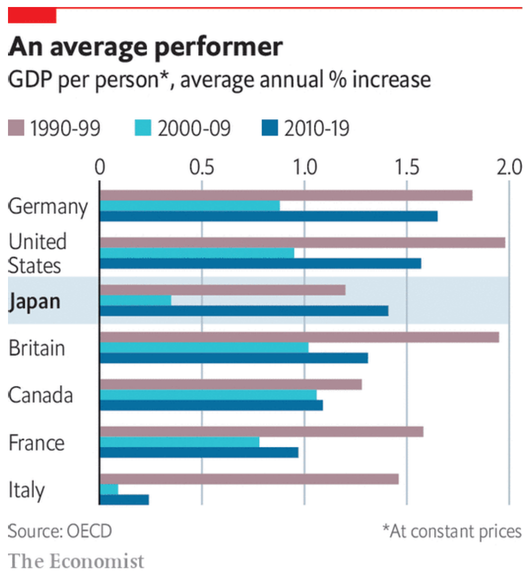
Seeking fiscal space

Yet some reckon fiscal policy ought to be used more forcefully. Many now regret two ill-timed consumption-tax increases in 2014 and 2019. Mr Abe's preferred candidate in the recent LDP leadership race called for postponing the government's primary-balance target until the BOJ hits its inflation target. In late November Mr Kishida's government announced a huge fiscal stimulus worth ¥55.7trn ($483bn).

Getting inflation up has not proved easy. Under Mr Kuroda, the BOJ expanded quantitative easing, adopted inflation targeting, and purchased a wider variety of assets. That helped pull the country out of its mild deflation, but only barely. "We misunderstood the inflation issue," says Mr Posen. "It turns out that secular stagnation is far more real and persistent than we thought."

Once inflation expectations are anchored around zero, raising them is hard. Moreover wages have not risen much, despite a tight labour market. Unions prefer stability of employment to wage rises, reckons Nakaso Hiroshi, a former deputy governor of the BOJ. Companies have gradually taken on more "non-regular" workers on part-time contracts; they account for 40% of the labour force, twice as much as in 1990. Perverse incentives may help depress their wages: many women limit their hours or incomes to secure a tax deduction for married couples earning below a threshold.

Figure C.2



Ageing, shrinking populations may also be weighing on demand and thus inflation. For Mr Shirakawa, that is a vindication of sorts. He argued at the BOJ that deflation was more a symptom of factors causing low growth, not the cause. In a new book, "Tumultuous Times", Mr Shirakawa says the impact of demographic change on growth "is still under-appreciated".

Overall growth has remained sluggish, but growth per head has recently been comparable with others in the G7. Unemployment has been minimal, longevity has increased and inequality has stayed relatively low. "Maybe Western economists who were so critical of Japan circa 2000, myself included, should go to Tokyo and apologise to the emperor," Paul Krugman, an economist, tweeted in 2020. "Not that they did great; but we did much worse."

Yet Japan could still do better. Public spending should be aimed more at improving long-term growth. Economists in Tokyo fret that the government has wasted its pandemic stimulus on handouts: a study by Hoshi Takeo of the University of Tokyo finds that fiscal support in 2020 was more likely to go to companies that were already struggling before covid-19.

Boosting productivity could help to offset the impact of the shrinking population. Mr Yoshikawa reckons that innovation is key to growth, and that ageing creates new problems that entrepreneurs can solve. Generational shifts may help. While many still prefer stable sarariman (salaryman) jobs in big firms, some of today's brightest graduates go into startups. "There is a burgeoning wave–they are a different species," says Mr Niinami. But structural reforms are necessary too, particularly to the inflexible labour market. It should be easier for workers to move between firms and industries, and harder for firms to exploit non-regular workers.

Yet pressure for reforms is lacking, partly due to resistance from vested interests, but also because life remains comfortable enough. "It's not an acute disease, it's chronic," says Mr Nakaso. "You don't really feel the pain, but it impacts your health in the long-term." This could be treated. As Mr Posen puts it, "There are 10,000 yen notes lying on the ground waiting to be picked up." Will Japan's leaders grab them?